UNIT-2

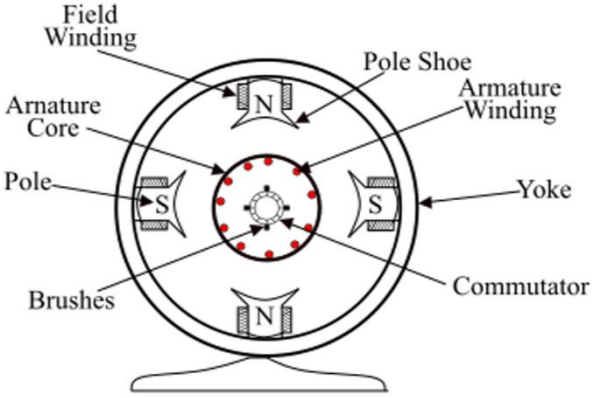
Machines and Measuring Instruments:

Principles and operation of DC machines, Transformers – Synchronous Machines - three Phase and single-phase induction motors - Moving coil and moving iron instruments, Wheatstone bridge and Megger.

1. Explain the principle of operation of DC generator Construction

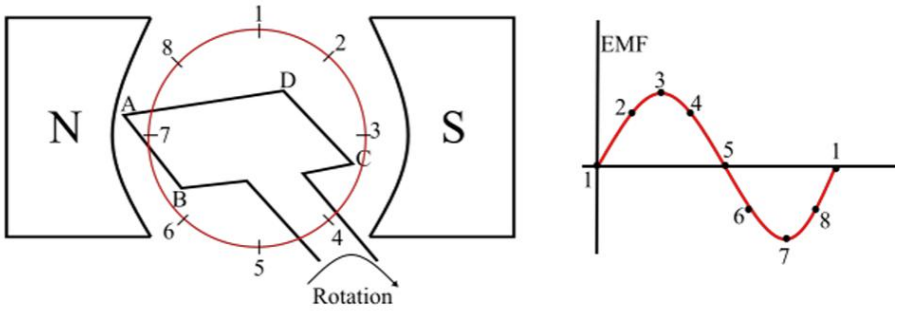
The components of a DC generator are,

1. Stator (Yoke, Poles, Field winding)
2. Rotor (Armature core & Armature windings)
3. Commutators
4. Brushes



Working Operation:

A DC generator operates on the **principle of electromagnetic induction** i.e. when the magnetic flux linking a conductor changes, an EMF is induced in the conductor. The EMF induced in the armature winding is AC (alternating current) and is converted into DC (direct current) using a commutator mounted on the rotor shaft of the generator.



Consider a single loop DC generator (as shown in the figure), a rectangular coil 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.

- When the loop is in position-1, the generated **EMF is zero** because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a **small EMF** is generated.

- When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated **EMF is maximum**.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a **reduced EMF** is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, **no EMF** is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

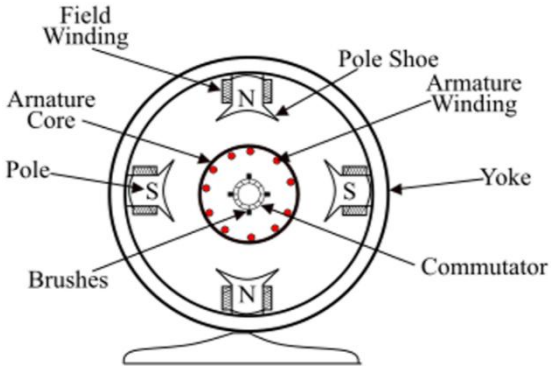
The generated EMF in the loop is **alternating voltage**. It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole. By using a commutator, this alternating emf generated in the loop can be converted into direct voltage.

2. Explain the operation of DC motor with neat diagram

Construction

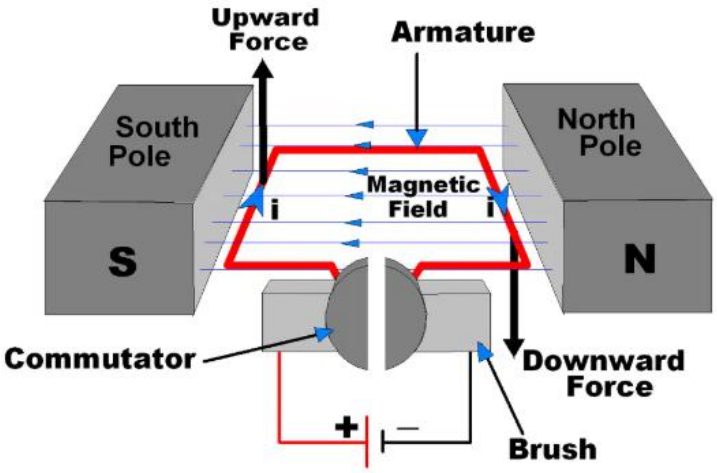
The components of a DC motor are,

1. Stator (Yoke, Poles, Field winding)
2. Rotor (Armature core & Armature windings)
3. Commutators
4. Brushes



Working Operation:

The DC motor working principle is that a when a current-carrying conductor placed in a magnetic field it experiences a mechanical force. This is known as the **Lorentz force.** The direction of this force is given by Fleming's Left-Hand Rule.



Consider a coil carrying DC current through a commutator and brushes. These commutator segments rotate freely about their axis. The commutator segment which contacts with the left brush gets positive polarity while the right one gets negative polarity. This leads to the flow of current in the coil.

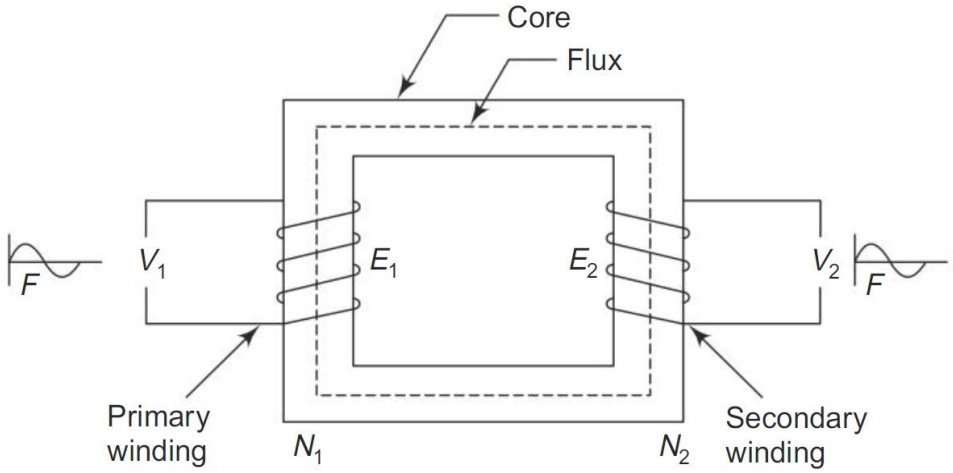
By applying Fleming's Left-Hand Rule, the conductor on the left side always experiences a force in an upward direction while the conductor on the right side experiences a downward force. Hence, a unidirectional torque is achieved in DC motors.

Back EMF

The interaction of the current-carrying conductor with the changing magnetic field produced by the field winding induces an EMF in the conductor. This EMF acts in the opposite direction to the applied voltage. This induced EMF in the motor is known as Back EMF.

3. Explain the operation of transformer with neat diagram.

Transformer is a static electrical machine which transforms electrical power from one circuit to another circuit, without changing the frequency. A transformer works on the principle of **electromagnetic induction** to step up or step down the voltage.



Parts of Transformer:

A transformer majorly consists of three parts,

1. **Magnetic Core**
   It is made up of laminated silicon steel, and it acts as magnetic flux flow channel.
2. **Primary winding**
   The Primary winding is the copper wires that are wound over the magnetic core, and it is connected to the input power source
3. **Secondary winding**
   The windings from which output is derived is known as secondary winding.

Working Operation:

- The transformer works on the principle of Mutual induction between two magnetically coupled coils is based on Faraday's laws of Electromagnetic Induction
- When primary winding is connected to AC supply mains a current flow through it
- This current produces an alternating flux in the core
- This flux links with primary winding and produces self-induced EMF in the primary winding which opposes the applied voltage
- This flux passes through the core and links with the secondary windings to induce an E.M.F called mutually induced E.M.F in the secondary winding
- The frequency of the EMF induced in the secondary is same as that of the flux or that of the supply voltage.
- The induced EMF in the secondary winding will be able to circulate the current in the external load connected to it
- Energy is transferred from primary winding to the secondary winding by electro-magnetic induction principle without change of frequency.

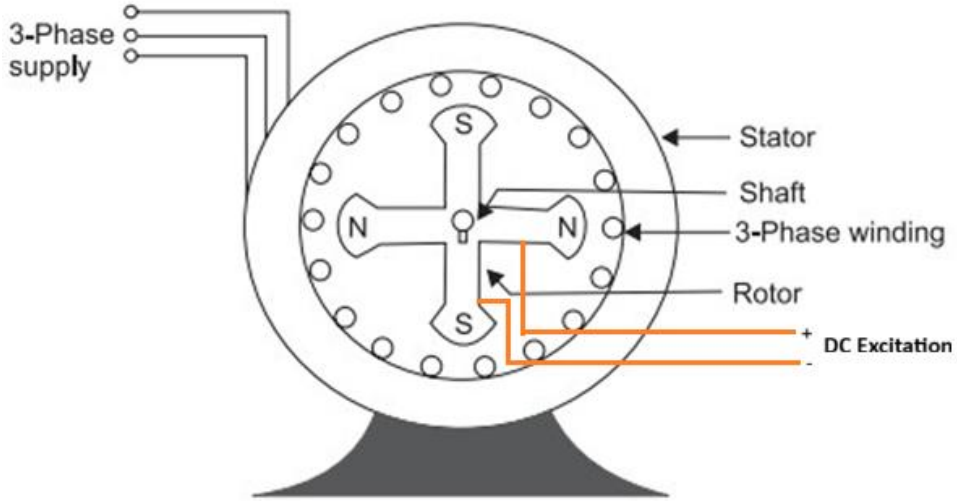
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4. Explain the operation of Alternator or synchronous Generator

An alternator (or synchronous generator) converts mechanical energy to electrical energy in the form of AC (alternating current). The term "synchronous" indicates that the rotor's rotational speed is directly proportional to the frequency of the generated voltage.

Main Components

- **Stator:** Contains three-phase or single-phase windings where the electrical current is induced.
- **Rotor:** It creates the magnetic field needed for induction in the stator using an excitation system.
- **Excitation System:** Supplies the DC direct current necessary to magnetize the rotor.



Working Operation:

The working principle of alternator is based on the **Faraday's law of electromagnetic induction**.

- The rotor winding is energized from the DC exciter and hence alternatively N and S poles are developed in the rotor.
- When the rotor rotates, its magnetic field cuts the **three-phase stator windings**, and an **alternating emf** is induced in each phase.
- The direction of induced e.m.f. can be found by Fleming right-hand rule.
- Because the rotor poles are N and S alternately, the direction of induced emf keeps changing, producing a sinusoidal AC voltage.
- When the rotor reaches a certain speed, the rotating magnetic field of the rotor tends to **lock** with the stator's rotating magnetic field.
- This condition is called **magnetic locking**, because of this locking, the rotor runs exactly at **synchronous speed**, and the **frequency** of the generated emf remains constant.
- **Synchronous speed** is given by

$$Ns = \frac{120f}{P}$$

where f= frequency of the generated e.m.f.,

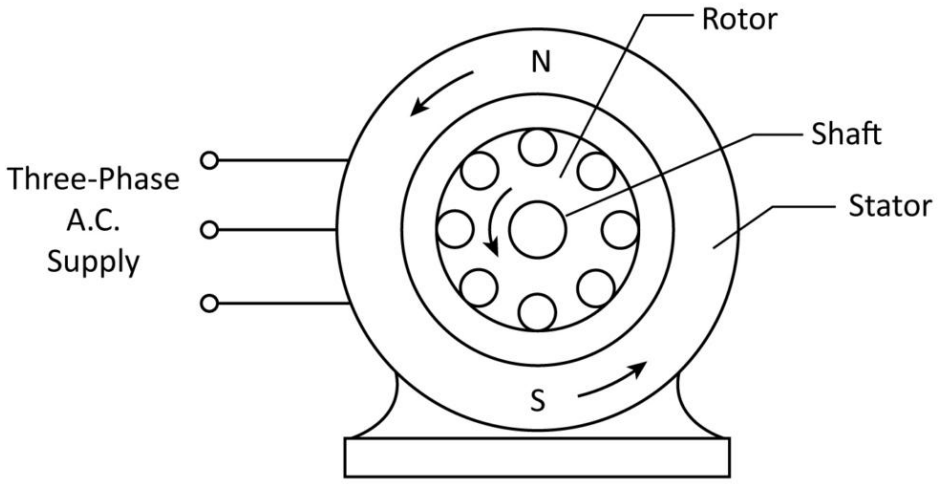
P= No.of poles

5. Explain the operation of Three Phase Induction motor

✓ A 3-phase induction motor is a singly-excited a.c. machine.

- Only stator is given 3-phase a.c. supply.
- The rotor is not given any supply.

An induction motor **runs at a speed other than the synchronous speed**. So, it is also known as asynchronous motor.



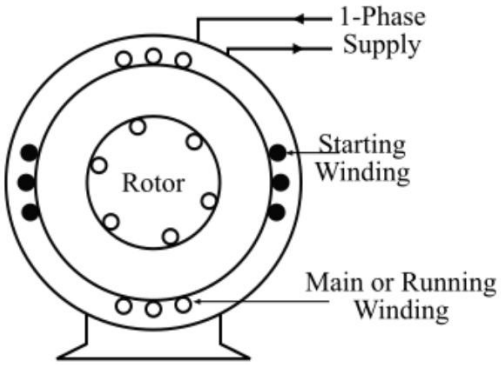
Operating Principle of a 3 phase Induction Motor

- A 3-phase induction motor works on the principle of **electromagnetic induction.**
- When a 3-phase a.c. supply is given to 3-phase stator winding, a **rotating magnetic field** (RMF) is created.
- The RMF is of constant magnitude and rotates at synchronous speed (N_s =120f/P)
- This rotating magnetic field induces emf in the rotor windings.
- Since the rotor is a closed (short-circuited) circuit, the induced e.m.f. causes a **current to flow** in the rotor.
- As per **Lenz's law**, the induced current and induced flux of the rotor act in such a direction as to oppose the very cause producing it.
    - The **effect** is the: Current in the rotor.
    - The **cause** is the: flux cutting by the rotating magnetic field.
- Here, the cause to be opposed is the **relative speed** between RMF and rotor.
- To reduce the relative speed (flux cutting), the rotor starts rotating in the **same direction** of the RMF.
- Induction motor rotor always rotate at a speed less than synchronous speed The difference between the flux (N_s) and the rotor speed (N_r) is called slip.

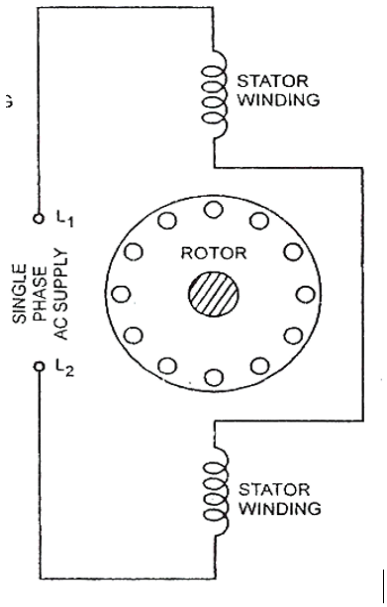
$$\% \text{ slip} = \frac{N_S - N_r}{N_S}$$

6. Explain the operation of Single-Phase Induction motors

A single-phase induction motor operates on the principle of **electromagnetic induction**, converting electrical energy from a single-phase AC supply into mechanical energy.



(OR)



Working Operation:

- A single-phase induction motor works on the principle of **electromagnetic induction**.
- When a single-phase AC supply is given to the stator, it produces an **alternating magnetic field**.
- According to the Double Revolving Field Theory, this field acts like two equal **rotating fields moving in opposite directions**. These fields induce emf and current in the rotor, and their interaction produces torque. But at standstill, the torques cancel each other, so the motor is **not self-starting**.
- To start the motor, the stator is provided with a **main winding** and an **auxiliary (starting) winding**, placed 90° apart. The auxiliary winding, usually with a capacitor or higher resistance, creates a **phase difference** between the two winding currents. This phase difference produces a **rotating magnetic field** during starting, giving the motor the required **starting torque**.

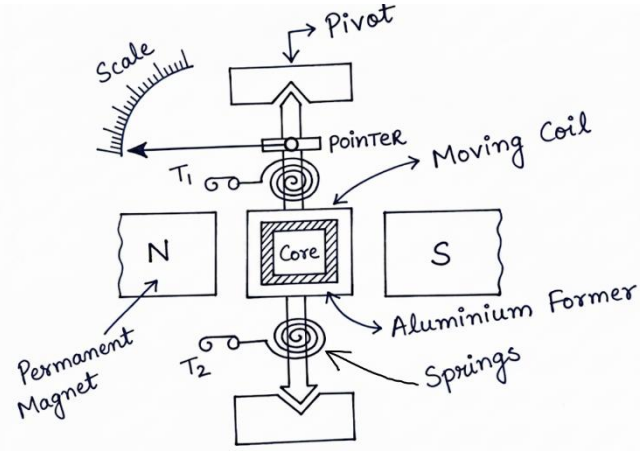
- Once the rotor starts turning and reaches around 70–80% of its speed, the auxiliary winding may be disconnected (depending on motor type). After that, the motor continues to run using only the main winding. During running, the forward rotating field becomes stronger, and continuous torque is produced by the interaction of this field with the rotor currents.

7. Explain the working of PMMC instrument with neat diagram

A **Permanent Magnet Moving Coil (PMMC)** instrument works on the principle that when current flows through a coil placed in a magnetic field, a torque is produced**.** If the permanent magnet is used in the instrument for creating the stationary magnetic field in which current-carrying coil moves, it is known as the permanent magnet moving-coil instrument.

Components

- Moving-Coil
- Magnet System
- Control Spring
- Damping
- Pointer and Scale.



Working Operation

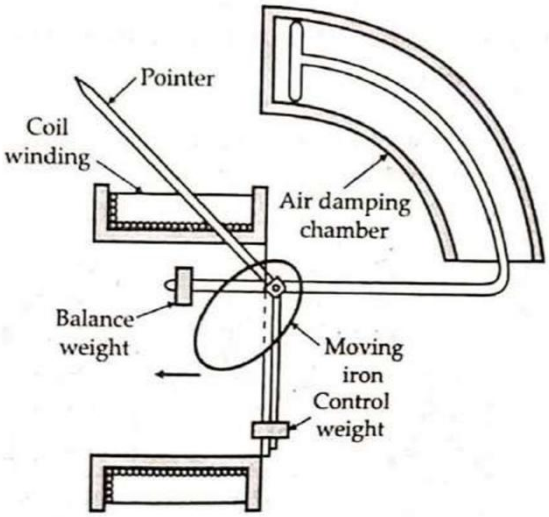
- In a PMMC instrument, a light rectangular coil (moving coil) is placed between the poles of a strong permanent magnet. When DC current flows through this coil, it experiences a force that makes it rotate.
- The deflecting torque produced is **proportional to the current flowing in the coil**, so the pointer moves over a scale and shows the value of the current.
- A **spring** provides controlling torque, which helps bring the pointer to rest at the correct position.
- A light aluminium frame attached to the coil provides **damping** due to eddy currents, making the pointer movement smooth and steady. Thus, the pointer takes a position where the **deflecting torque equals the controlling torque**, giving a linear and accurate reading.

8. Explain the working of attraction type moving iron instrument

A moving iron piece is utilized for calculating either the current or voltage flow is termed as the moving iron instrument.

Construction:

- It consists of a fixed coil and moving iron piece.
- The moving iron piece is a flat disc which is mounted on the spindle.
- The controlling torque is provided by either spring control or gravity control.
- Here the damping torque is provided by the Air friction damping.



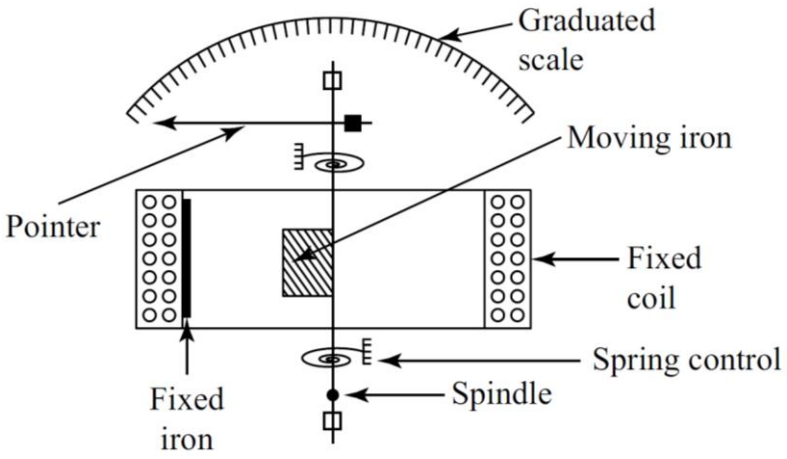
Working Operation:

- An attraction-type moving iron instrument works on the principle that **a soft iron piece becomes magnetised and is attracted toward a stronger magnetic field**.
- When current flows through the **fixed coil**, it produces a magnetic field. Inside this coil, a small **soft iron piece** is placed. As the magnetic field becomes stronger with the increase in current, the soft iron piece gets magnetised and is **pulled (attracted)** toward the region of maximum magnetic field.
- This movement of the iron piece causes the **pointer** attached to it to move over the scale. The more the current through the coil, the stronger the magnetic field, and the greater the attraction and pointer deflection.
- A **controlling spring** brings the pointer to its final steady position, and **air damping** provides smooth movement.

9. Explain the working of repulsion type moving iron instrument

Construction:

- Repulsion type instrument consists of a fixed coil which is wound over the hollow cylinder and two iron pieces that are,
    - **Fixed iron piece**
    - **Movable iron piece**
- Fixed iron piece is attached to the hollow cylinder and moving iron piece is attached to the spindle.
- Controlling torque is provided by spring control and damping torque is provided by Air friction damping.



Working Operation:

- A repulsion-type moving iron instrument works on the principle that two similarly magnetised soft-iron pieces repel each other when current flows through the coil.
- Inside the fixed coil, **two soft iron vanes (pieces)** are placed very close to each other.
- When current passes through the coil, both iron pieces get **magnetised with the same polarity**. Because they now have like poles, they **repel** each other.
- One iron piece is fixed, and the other is free to move. Due to the repulsive force, the moving iron piece rotates and carries the **pointer** across the scale.
- As the current increases, the magnetic field increases, the iron pieces become more strongly magnetised, and the **repulsion force increases**, causing greater pointer deflection.
- A **controlling spring** provides restoring torque and helps the pointer settle at the correct position.
- Since the repulsive force depends on the **square of the current (I^2)**, the instrument can measure **both AC and DC**. The scale is **non-uniform**.

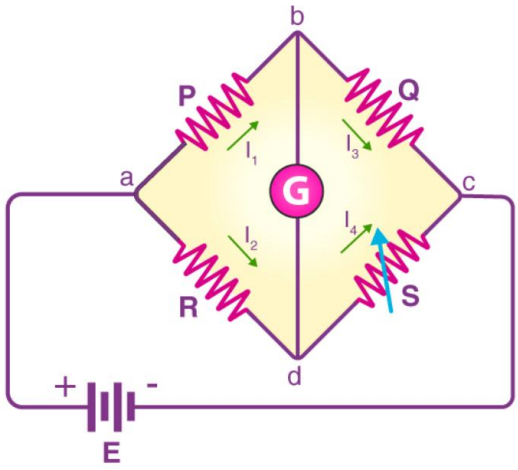
10.Explain the working of Wheatstone bridge with neat diagram

- The Wheatstone bridge works on the principle of null deflection, i.e. the ratio of their resistances are equal and no current flows through the circuit.
- It is used to measure an unknown resistance by comparing it with known resistances.

The bridge has four resistances arranged in the form of a diamond:

- Two known resistors
- One variable resistor
- One unknown resistor

A galvanometer is connected between the middle points, and a DC supply is connected across the other two points.



Working Operation

The bridge consists of **four resistors** arranged in a diamond shape:

- P and Q form one pair of opposite arms.
- R and S form the other pair.
- A galvanometer (G) is connected between the junction of P–Q and R–S.
- A DC supply is connected across the other two opposite junctions.

When the supply voltage is applied, currents flow through the two branches **(P–Q branch and R–S branch)**. The galvanometer detects the **difference in potential** between the midpoints of the two branches.

Balance Condition

The bridge is said to be balanced when no current flows through the galvanometer. This occurs when the potential drop across P is proportional to that across Q, and similarly R to S.

The balance condition is:

$$\frac{P}{Q} = \frac{R}{S}$$

The unknown resistor R can be found by

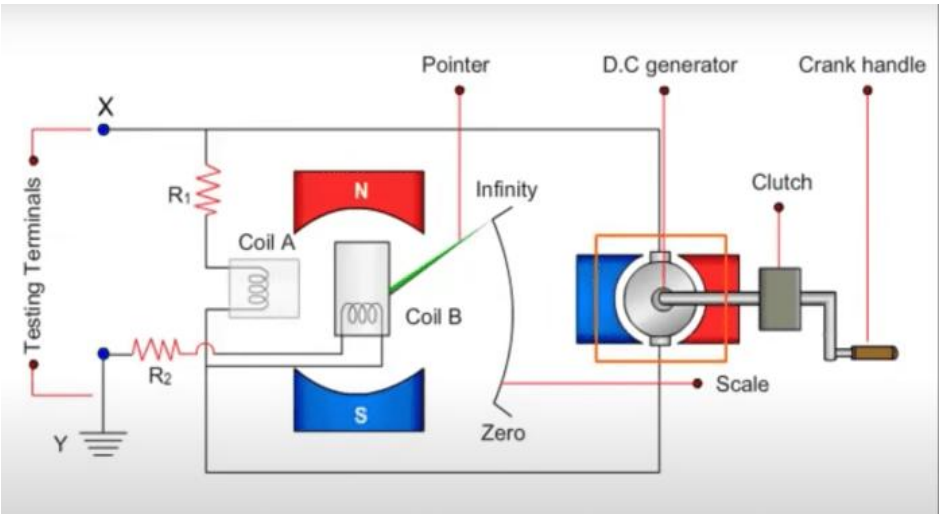
$$R = \frac{P}{Q} \times S$$

Where,

- R is the unknown resistance
- S is the known resistance of standard arm
- P and Q is the ratio of the arm of the bridge

11. Explain the working of Megger with neat diagram

Megger is a device used to measure insulation resistance. Megger is a combination of both generator and an ohmmeter. This instrument is used to measure the high-value resistance.



Megger consists of the following parts.

1. DC generator
2. 2 Coils (Coil A, Coil B)
3. Clutch
4. Crank handle
5. terminal X & Y

Working Principle:

Megger works on the principle of **Electromagnetic Induction**. The operation of a megger is based on a **moving coil** meter. When current is flowing in a conductor, which is placed in a magnetic field, it experiences a torque.

- When the crank handle is rotated, the **clutch connects the handle to the DC generator**, so the generator starts producing **high DC voltage**.
- This voltage is applied to the insulation under test through **terminals X and Y**.
- **Coil A (Deflecting coil)** carries the **leakage current** that flows through the insulation.
- **Coil B (Controlling coil)** carries a current proportional to the **generated voltage**. Both coils produce **torques in opposite directions**.
- If the insulation is **good**, leakage current is very small → Coil A produces less torque → pointer moves to the **HIGH resistance** side.
- If the insulation is **bad**, more leakage current flows → Coil A produces more torque → pointer moves to the **LOW resistance** side.
- The pointer finally settles when the torques from Coil A and Coil B become equal. This balanced position directly shows the insulation resistance in **megaohms** on the scale.

All the best